# Mental overload is invisible until something goes wrong

A live, personal load reading from a consumer EEG headset, cleaned well enough to be trusted

## Surgeons, pilots, controllers and shift workers all overload quietly.

Mistakes follow the overload; the overload itself has no gauge.

## EEG carries the signal: alpha falls and theta rises when the mind is working hard.

Measured in 36 lab people, it shows up within seconds.

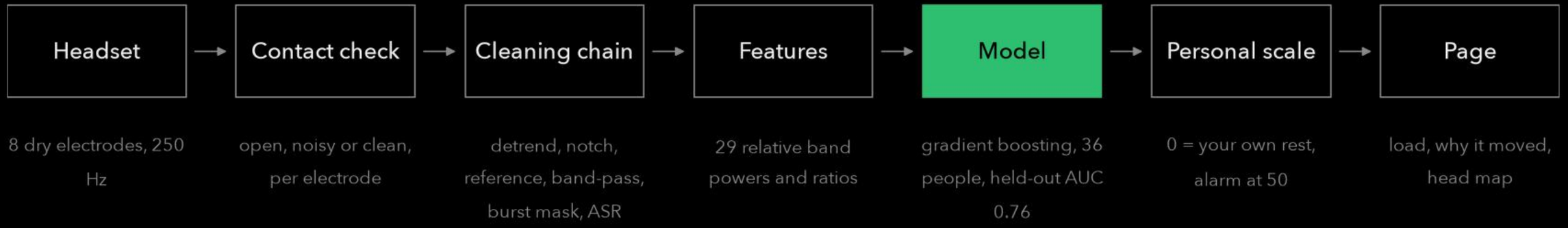
## The problem is the hardware. Dry electrodes on hair give 6,000 µV of noise around 30 µV of brain.

So the real work is cleaning the signal, and refusing to score it when it cannot be cleaned.

Source: PhysioNet EEG during mental arithmetic (36 people); recordings on a g.tec Unicorn BCI Core-8, 7 and 8 Nov 2026.

# Eight electrodes become one number, once a second

Every stage runs on the last 4 seconds of EEG; the model is the only learned part



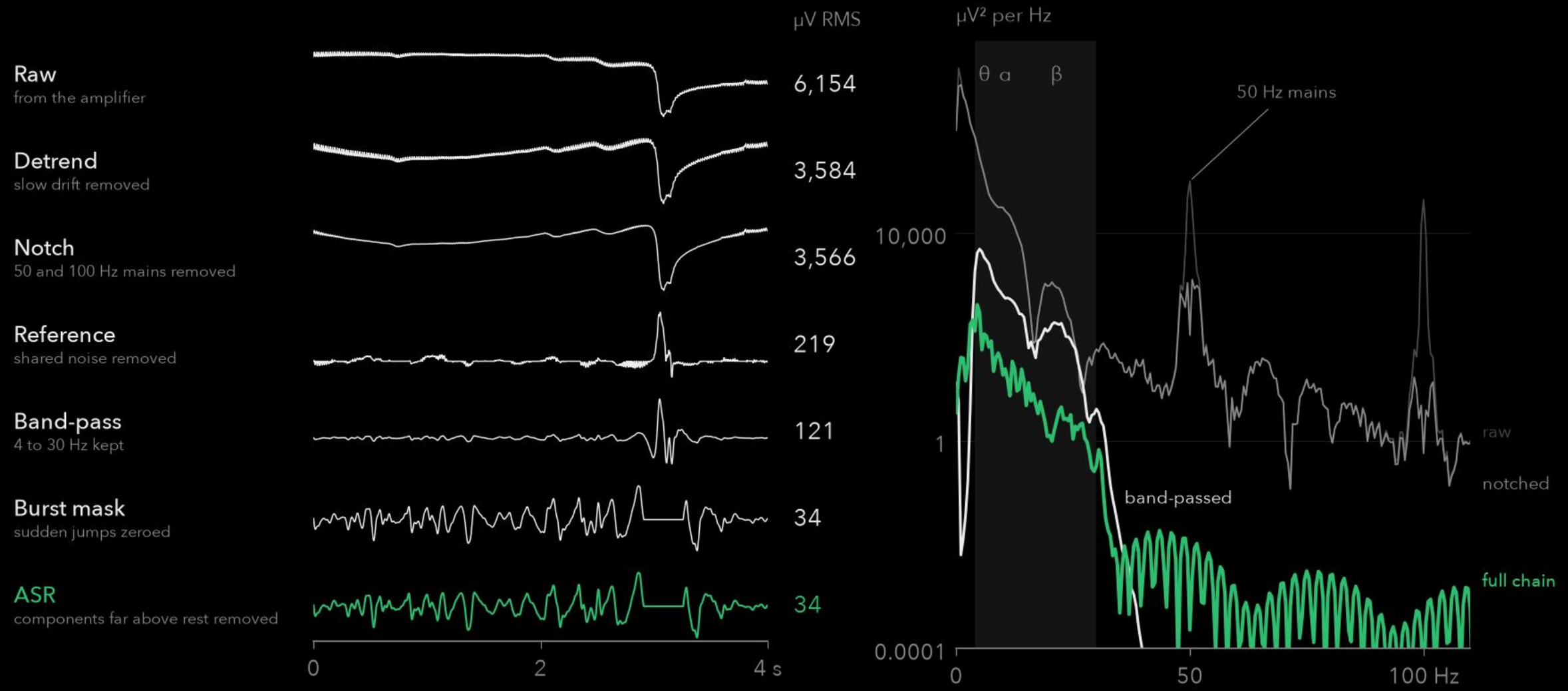

If any model electrode is open the number is held; if one is noisy the number is shown grey with the alarm off.

A bad signal is reported as bad, never scored.

Source: loadmonitor/load_gauge.py, gauge_server.py, web/index.html. Optional Claude debrief when an API key is set.

# Six filters take a dry electrode from 6,000 µV of noise to 34 µV of EEG

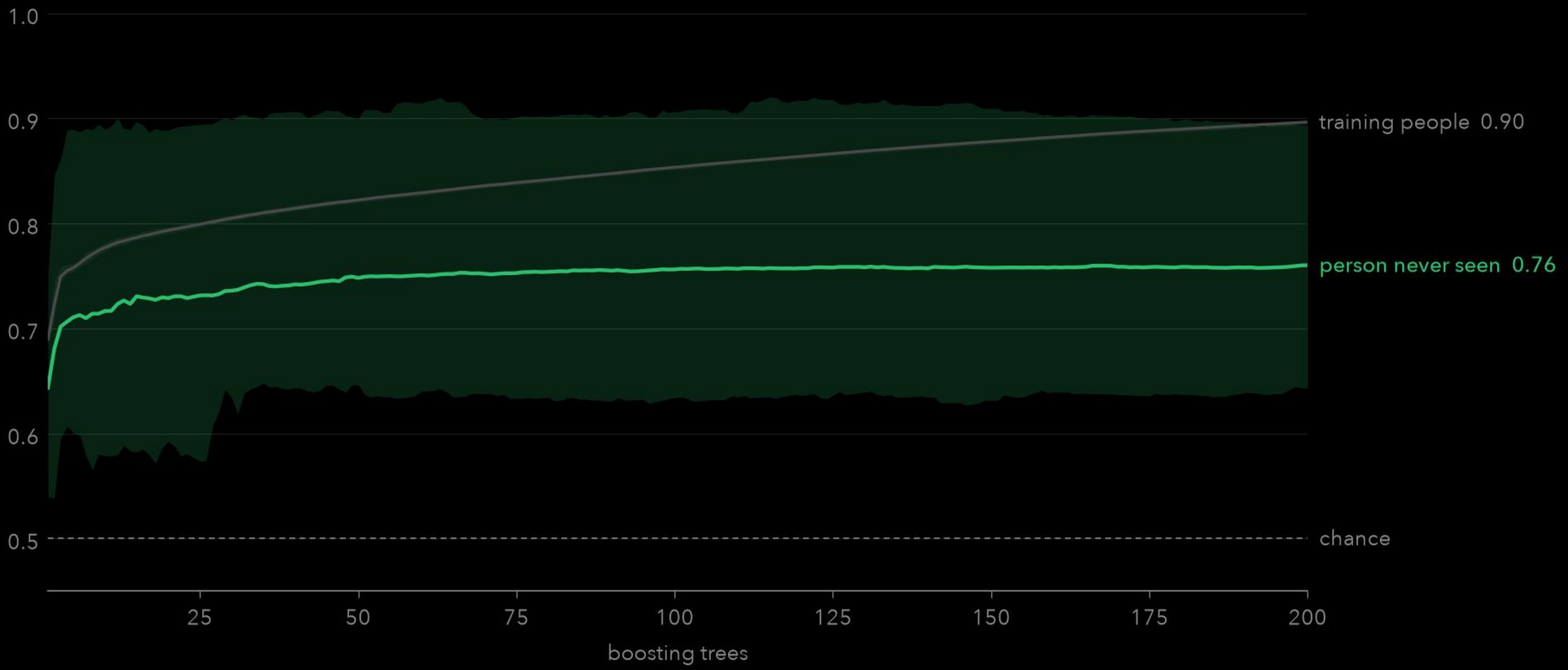
The same 4 seconds at Pz on Paing's cap: the signal after each stage (each row on its own scale) and its power spectrum




Source: recording paing_20261108_011525, Pz. Stage outputs verified identical to the live code (model_search.clean_window). Scalp EEG is about 10 to 50 µV. Spectrum: Welch, 2 s segments.

# On a new person the model stops improving after about 50 trees

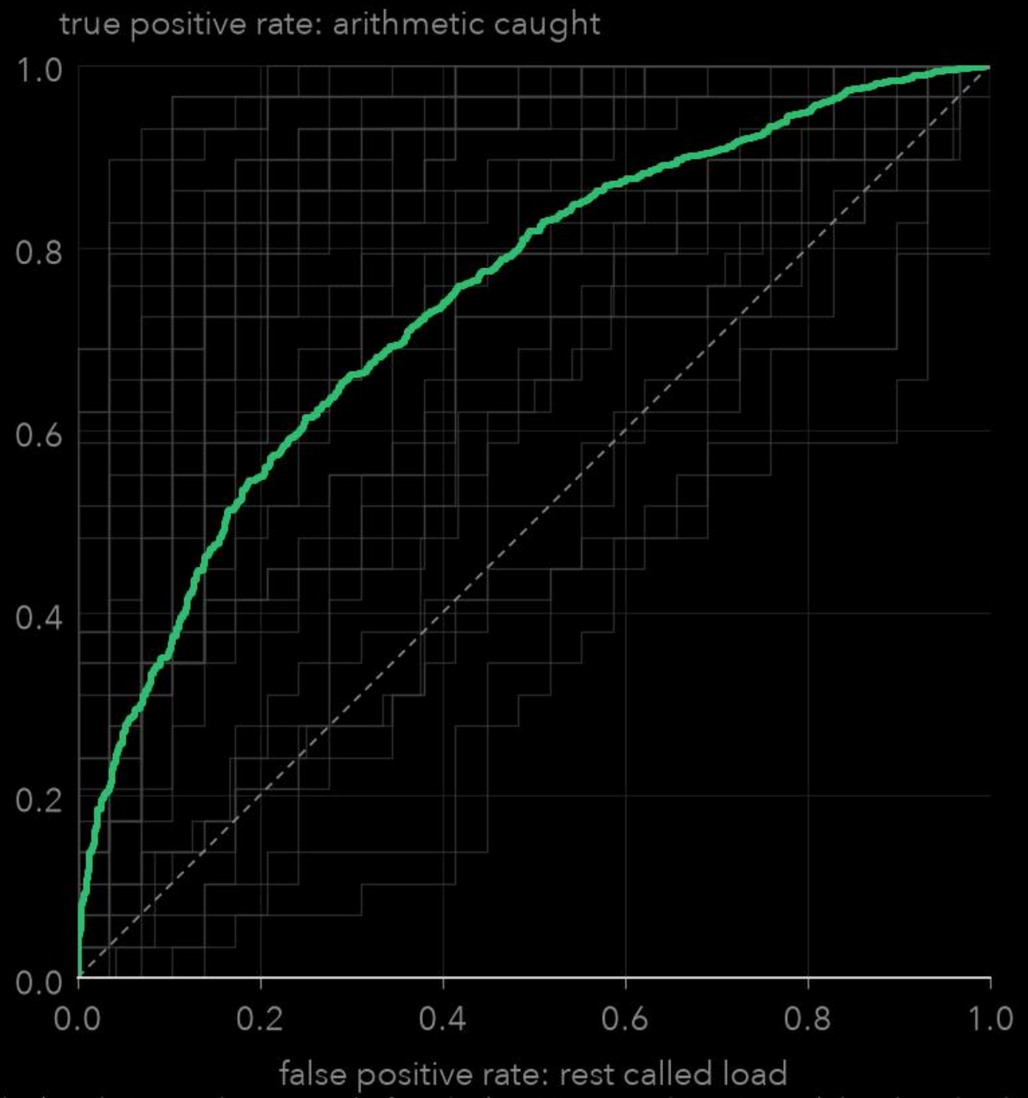
AUC, rest vs arithmetic, as trees are added; mean of 36 leave-one-person-out rounds, band = middle half




Source: PhysioNet EEG during mental arithmetic, 36 people. Live recipe: 5 channels, cleaning chain, 29 amplitude-invariant features, 200 trees of depth 2.

# On people it has never seen, the model reaches an AUROC of 0.74

Each of 36 people scored by a model trained on the other 35, one 4 s window at a time; grey = one person, green = all pooled




**0.74**
pooled AUROC, held out

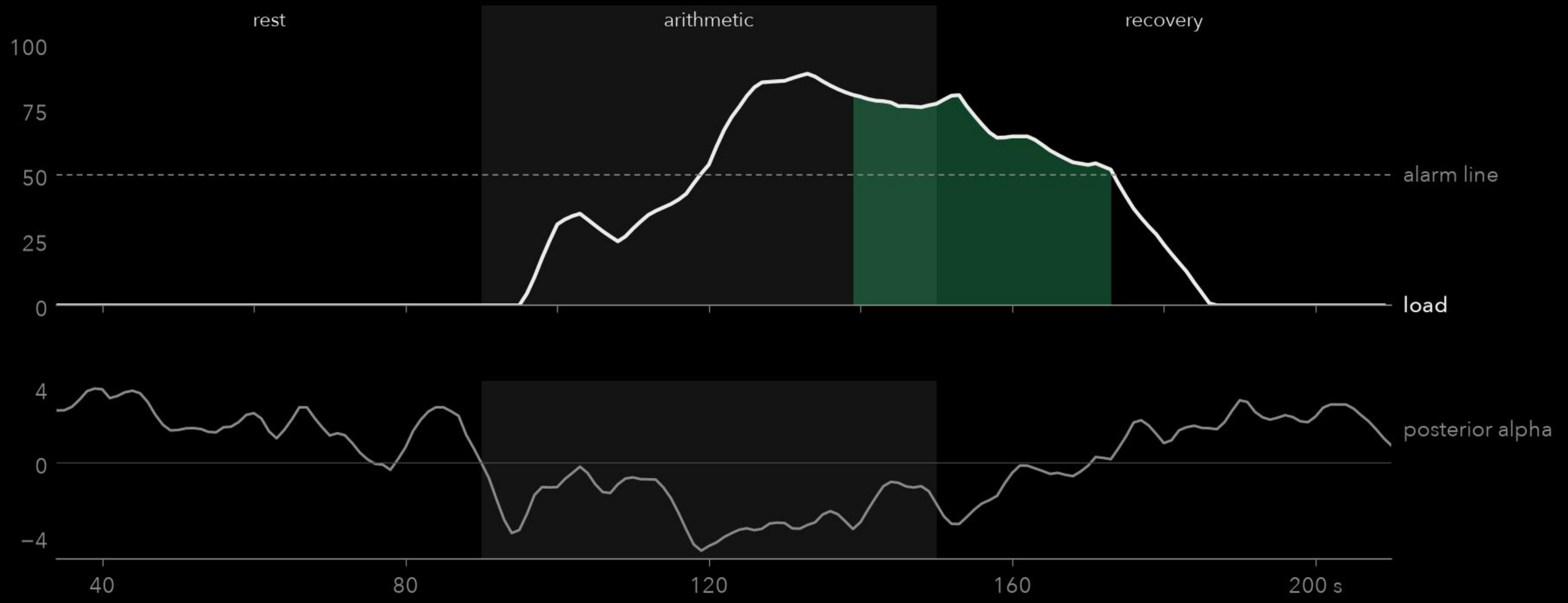
**0.81**
median per person

**23 of 36**
people at 0.7 or better

Source: PhysioNet EEG during mental arithmetic (gel cap). Per-window scores before the live 12 s smoothing. Not validated on the dry cap.

# A held-out person's load climbs during arithmetic and falls back at rest

Lab EEG replayed through the live gauge; load with 0 = that person's own rest; posterior alpha below, dB vs rest




Source: PhysioNet eegmat person 09, held out of the model's training. Shaded span = where the alarm was on (load above 50 for 20 s; it lingers from the 12 s smoothing).

The recovery minute is the same person's earlier rest; the dataset has no rest after the task. Alpha: 8 to 13 Hz, mean of Pz and O1, 9 s smoothing.

MENTAL LOAD eeg · 8 ch · 250 hz

BASELINE | MENTAL | PHYSICAL | RECOVERY | NONE

MENTAL ■ RUNNING 02:29

[1] LOAD · RELATIVE TO YOUR BASELINE — ALARM > 50 FOR 20 S

# 77

HIGH LOAD alarm after 20 s above 50

Steady · alongside it: posterior alpha -3.7 dB, frontal theta -0.2 dB vs your rest

49

NEAR START · SLOW TREND, BUILDS OVER 20+ MIN

fatigue · (θ+α)/β change this session · 50 = your start · published ratio, not a trained model

[2] ELECTRODES · CHANGE FROM BASELINE — DB · SHOW CONTACT




LOAD HELD ABOVE THE ALARM LINE

[3] MARKERS

| | |
|---|---|
| Fz theta | -0.2 dB |
| Pz alpha | -4.4 dB |
| most active | O1 +-2.1 dB |
| dead | none |
| clean (under 100 µV after cleaning) | 5 / 5 |
| alpha at Pz/O1, 8 s median (lab rest ≈ +6 dB) | +5.4 dB |
| model | lab AUC 0.76, each of 36 people held out in turn · this copy trained without person 09 |

DEBRIEF | REBASELINE | ALPHA CHECK

[4] LOAD · LAST 10 MIN — 1 S




[5] SIGNALS · LAST 2 S — 1-40 HZ · MV



# From one headset to a gauge anyone can wear

Where this goes next

## A personal number, not a lab average.

Zero is your own rest, so the same gauge fits a surgeon on a night shift and a pilot on approach.

## Proven on people it had never seen.

Held-out AUROC 0.74 across 36 people; the pipeline runs end to end in real time on an eight-electrode consumer headset.

## Next: better electrodes, more people, longer sessions.

Then slow fatigue, which builds over 20 minutes or more, joins the live load on the same page.

Source: docs/OVERNIGHT.md in the repository, every number traceable to a script.